A Disconnected Counterexample to Edge-Addition Threshold Preservation for Positive Square Energy

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Abstract

For a graph G, let $s^+(G)$ be the sum of the squares of its positive adjacency eigenvalues. We give an exactly certified graph X and a nonedge e such that $s^+(X) > |V(X)|$ but $s^+(X+e) < |V(X)|$. Thus the natural edge-addition threshold-preservation statement is false without a connectedness hypothesis. The graph has order 607 and is a disjoint union of one nine-vertex graph and odd cycles. A short exact-arithmetic script certifies both strict inequalities by rational root isolation.

1 The counterexample

For a graph G with adjacency eigenvalues λ_i, put

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2.$$

Theorem 1. *There is a graph X of order 607 and a nonedge e such that*

$$s^+(X) > 607 > s^+(X+e).$$

In particular, the implication

$$s^+(G) \geq |V(G)| \quad \Longrightarrow \quad s^+(G+e) \geq |V(G)|$$

is false for disconnected graphs.

Proof. Let D be the nine-vertex graph with graph6 encoding

`HQzV]zn`,

using zero-based graph6 labels, and let $e = \{2, 3\}$. Direct decoding shows that e is a nonedge. Exact determinant calculation gives

$$\chi_D(x) = x^2(x+1)(x+2)^3(x^3 - 7x^2 + 6x + 6)$$

and

$$\chi_{D+e}(x) = x^2(x+2)^2(x^5 - 4x^4 - 14x^3 + 14x^2 + 18x - 6).$$

Exact rational root isolation, followed by squaring and summing only the positive root intervals, gives

$$36.665892147475496 < s^+(D) < 36.665892147475498$$

and

$$36.635149762774868 < s^+(D+e) < 36.635149762774870.$$

For $r \equiv 1 \pmod 4$, the cycle eigenvalues $2\cos(2\pi j/r)$ and the finite cosine-sum identity give

$$s^+(C_r) - r = 1 - \sec(\pi/r).$$

In particular,

$$s^+(C_5) - 5 = 2 - \sqrt{5}, \qquad s^+(C_{13}) - 13 = 1 - \sec(\pi/13).$$

Define

$$X = D \sqcup 117C_5 \sqcup C_{13}.$$

Then $|V(X)| = 9 + 117 \cdot 5 + 13 = 607$. Since positive square energy is additive over disjoint unions,

$$s^+(X) - 607 = s^+(D) - 9 + 117(2 - \sqrt{5}) + 1 - \sec(\pi/13),$$
$$s^+(X+e) - 607 = s^+(D+e) - 9 + 117(2 - \sqrt{5}) + 1 - \sec(\pi/13).$$

Exact outward rational interval arithmetic yields

$$0.016010949050374 < s^+(X) - 607 < 0.016010949050375$$

and

$$-0.014731435650253 < s^+(X+e) - 607 < -0.014731435650252.$$

Both required inequalities are strict. □

2 Exact certification

The accompanying script `disconnected_threshold_counterexample.py` performs the complete certificate. SymPy isolates every real root of the two integer characteristic polynomials in exact rational intervals of width below 10^{-35}. The bound on $\sqrt{5}$ is checked by rational squaring. To handle $\sec(\pi/13)$, the script eliminates $z = e^{i\pi/13}$ from

$$\Phi_{26}(z) = 0, \qquad y(z^2+1) - 2z = 0,$$

obtaining

$$y^6 + 6y^5 - 24y^4 - 32y^3 + 80y^2 + 32y - 64 = 0.$$

The root $y = \sec(\pi/13)$ is the unique root in $(1, 11/10)$; its inclusion there follows from the elementary bound $\cos x > 1 - x^2/2$ together with $\pi < 22/7$, which gives $\cos(\pi/13) > 10/11$. The script then evaluates both slacks using outward rational interval arithmetic and terminates only if the first lower bound is positive and the second upper bound is negative.

From the repository root, reproduce the certificate with

```
python positive-square-energy/experiments/\
  disconnected_threshold_counterexample.py
```

No floating-point value is used to establish either sign.

3 Scope

The graph X is disconnected. Therefore this result does not settle the connected edge-addition question and does not provide a counterexample to the connected positive-square-energy conjecture. It does show that any promotion argument from spanning bicyclic subgraphs must use connectedness or additional structure; threshold preservation is not a purely spectral statement valid for arbitrary disjoint unions.